Tutorial 9

Q1. Which of these collections of subsets are partitions of $\{1,2,3,4,5,6\}$

(a) $\{1,2\}, \{2,3,4\}, \{4,5,6\}$.

(b) $\{1\}, \{2,3,6\}, \{4\}, \{5\}$

(c) $\{2,4,6\}, \{1,3,5\}$.

(d) $\{1,4,5\}, \{2,6\}$

Q2. Which of these relations on $\{0,1,2,3\}$ are partial ordering? Determine the properties of the partial ordering that others lack?

(a) $\{(0,0), (1,1), (2,2) (3,3)\}$.

(b) $\{(0,0) (1,1) (2,0) (2,2) (2,3) (3,2) (3,3)\}$

(c) $\{(0,0) (1,1) (1,2) (2,2) (3,3)\}$.

(d) $\{(0,0) (1,1) (1,2) (1,3) (2,2) (2,3) (3,3)\}$

Q3. Is (S,R) a poset if S is the set of all people in the world and $(a,b) \in R$ where a, b are people if

(a) a is taller than b.

(b) a is not taller than b

(c) $a = b$ or a is ancestor of b

(d) a & b have common friend

Q4. Which of these are posets

(a) $(\mathbb{Z}, =)$

(b) $(\mathbb{Z}, \neq)$

(c) $(\mathbb{Z}, \geq)$

(d) $(\mathbb{Z}, \times)$

Q5. Determine which relations are posets

(a) $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}$

Q6. Find two incomparable elements in this poset

(a) $(P(\{0,1,2\}, \subseteq)$

(b) $(\{1,2,4,6,8\}, |)$

Q7. Draw the Hasse diagram for divisibility on the set

(a) $\{1,2,3,4,5,6,7,8\}$

(b) $\{1,2,3,5,7,11,13)$

(c) $\{1,2,3,6,12,24,36,48\}$

Q8 Answer these questions for the Poset $(\{3, 5, 9, 15, 24, 45\}, |)$.

(a) Find the maximal elements

(b) Find the minimal elements.

(c) Is there a greatest element

(d) Is there a least element

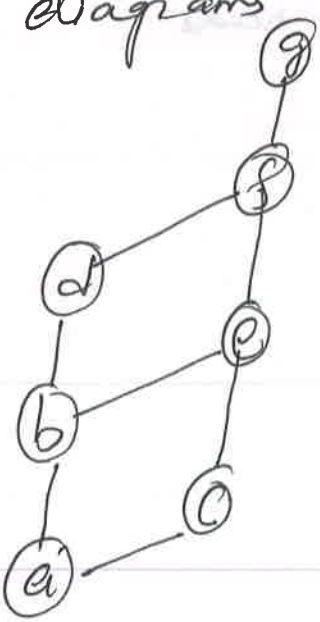
(e) Find all upper bounds of $\{3, 5\}$.

(f) Find the least upper bound of $\{3, 5\}$ if it exists

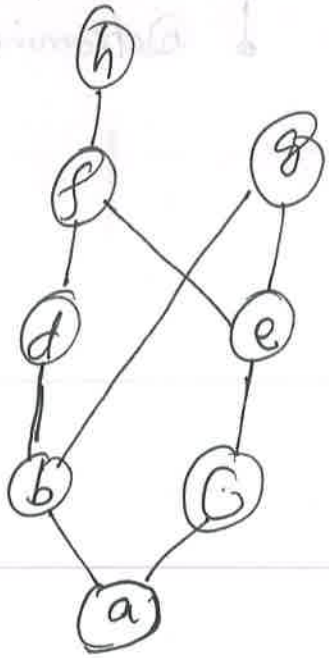
(g) Find all lower bounds of $\{15, 45\}$

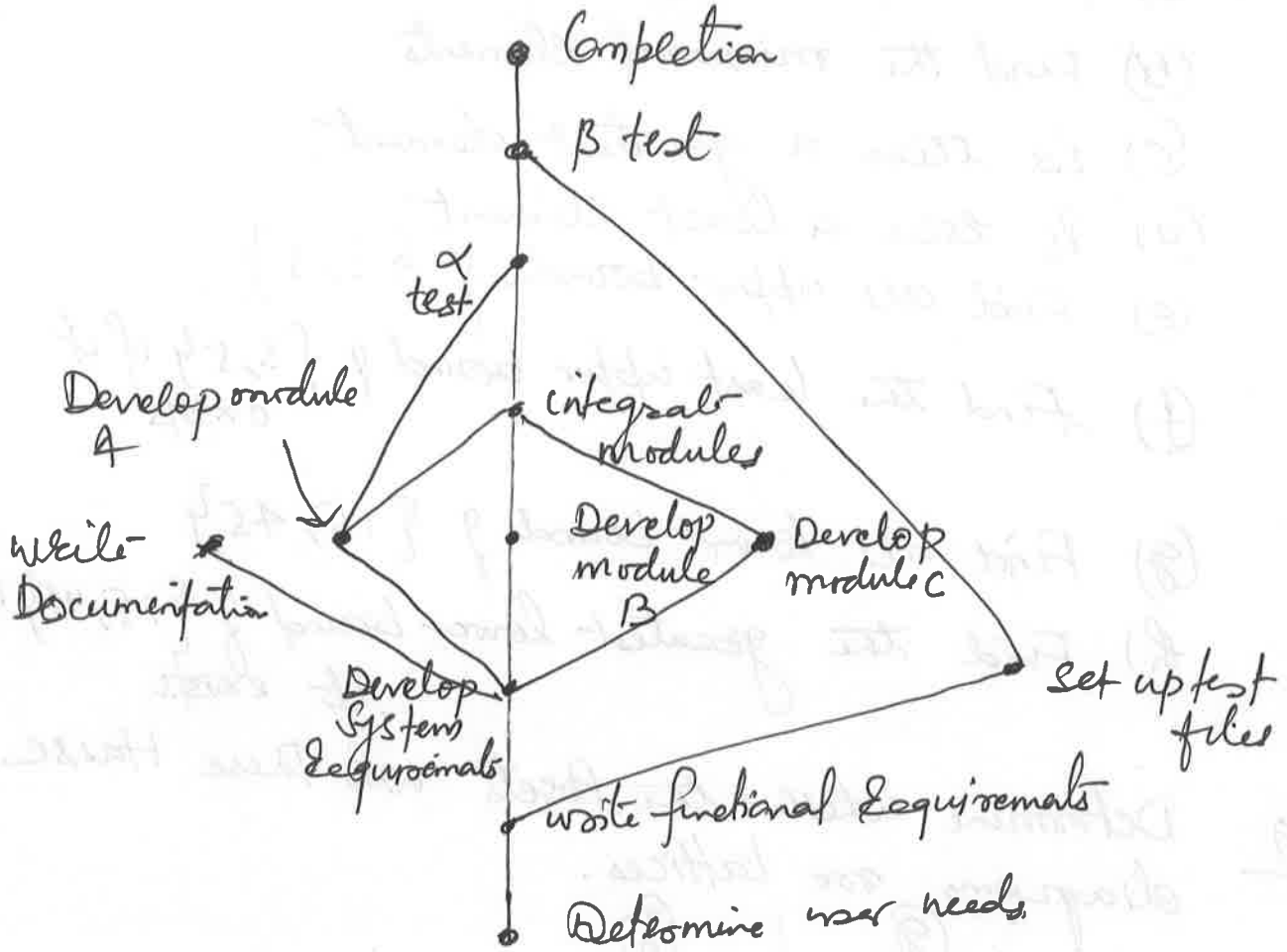
(h) Find the greatest lower bound of $\{15, 45\}$ if it exists.

Q9 Determine whether the Posets with these Hasse diagrams are lattices.



(a)



(b)

Q10. Find the ordering of the tasks of a software project if the Hasse diagram for the task of the project is shown